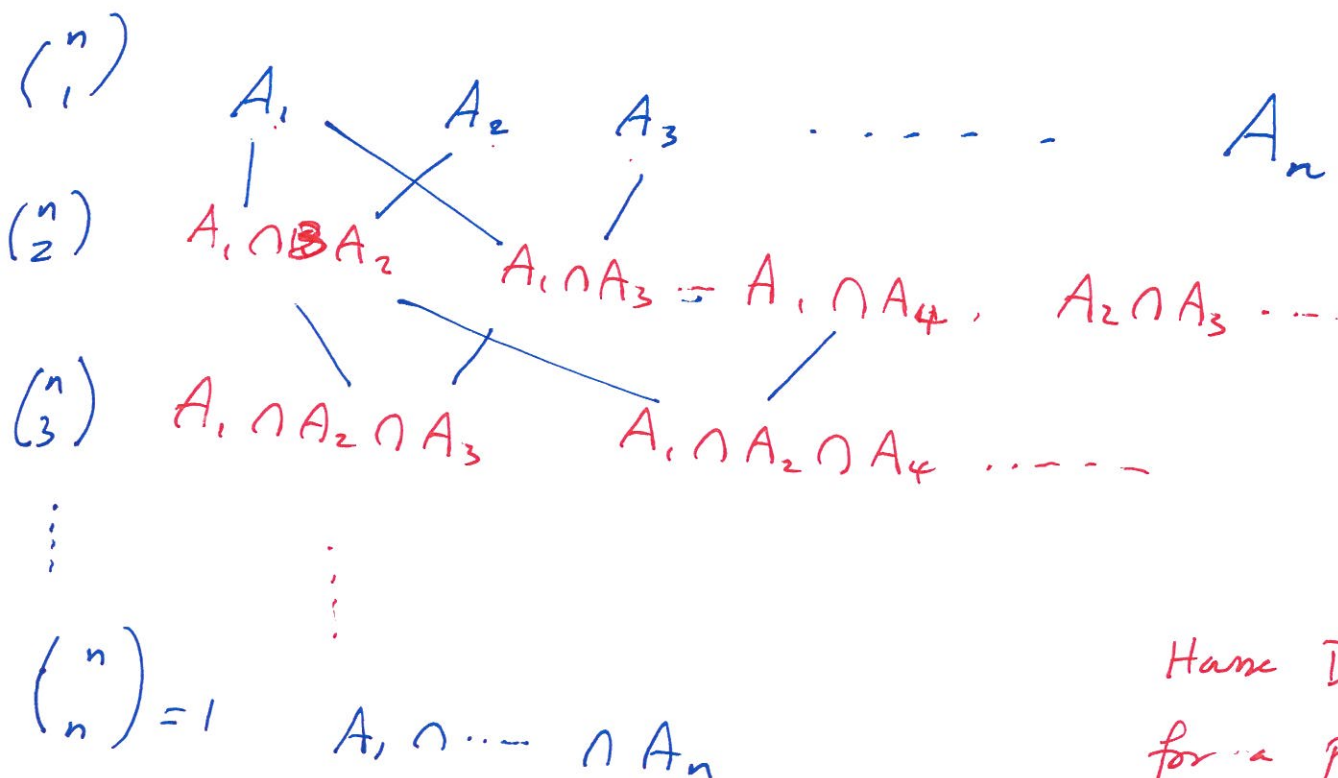


LECTURE 34: MATH1061/7861
TOPICS: INCLUSION EXCLUSION
DATE: TUESDAY MAY 19, 2026

Learning objectives:

- (1) Understand the Inclusion Exclusion principle for two sets and three sets.
- (2) Use the Inclusion Exclusion principle to solve counting problems.

Formula: $|A_1 \cup A_2 \cup \dots \cup A_n|$



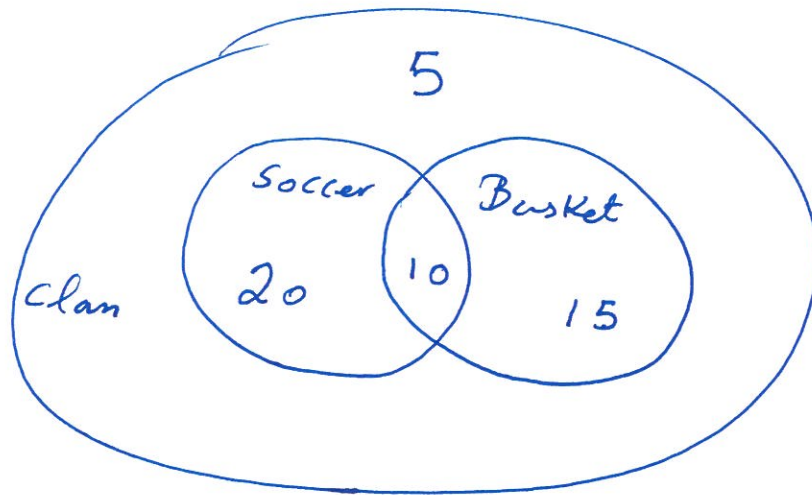
Hasse Diagram
for a partially
ordered set.

Challenge: Use induction to prove this

$$|A_1 \cup \dots \cup A_n| = |A_1| + \dots + |A_n| - |A_1 \cap A_2| - |A_1 \cap A_3| - \dots \\ + |A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| + \dots$$

Q1: In a high school class, 20 students play soccer, 15 play basketball, 10 play both and 5 play neither. How many students are in this class?

$$25 + 5 = 30$$



$A = \{ \text{students playing soccer} \}$

$B = \{ \text{students playing Basketball} \}$

$A \cap B = \{ \text{students playing both} \}$

$A \cup B = \{ \text{students who play soccer or basketball} \}$

$$|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 10 = 25$$

$(A \cup B)^c = \{ \text{students play neither} \}$

$$|(A \cup B)^c| = 5$$

Q2: In a database of 1000 users:

400 users use Chrome,
350 use Firefox,
300 use Safari,
150 use both Chrome and Firefox,
120 use both Chrome and Safari,
100 use both Firefox and Safari,
50 use all three browsers.

How many users use at least one of these browsers?

$$A = \{\text{users using Chrome}\}$$

$$B = \{\text{users using Firefox}\}$$

$C = \{ \text{users using Safari} \}$

We want $|A \cup B \cup C| = |A| + |B| + |C|$

$$= |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$= 400 + 350 + 300 - 150 - 120 - 100 + 50$$

$$= 730$$

Q2: In a school lunch room, students can bring chips, chocolate, or fruits.

- 20 brought chips A
- 25 brought chocolate B
- 15 brought fruits C
- 10 brought both chips and chocolate $A \cap B$
- 5 brought both chips and fruit $A \cap C$
- 7 brought both chocolate and fruits $B \cap C$
- 3 brought all three $A \cap B \cap C$

How many students are there (assuming everybody brings ~~some~~ thing)?

← at least one of these items.

Important: The last assumption implies

$$\# \text{students} = |A \cup B \cup C|.$$

$$= 20 + 25 + 15 - 10 - 5 - 7 + 3$$

$$= 41$$